



Sec: 5C-60

D-Block elements

Date: 24-06-2020

SINGLE ANSWER TYPE

- Identify the statement which is not correct regarding CuSO_4
A) It reacts with KI to give I_2
B) It reacts with KCl to give Cu_2Cl_2
C) It reacts with NaOH and glucose to give Cu_2O
D) It gives CuO on strong heating in air
- An aqueous solution of FeSO_4 , $\text{Al}_2(\text{SO}_4)_3$ and chrome alum is heated with excess of Na_2O_2 and filtered. The material obtained is
A) a colourless filtrate and a green residue
B) a yellow filtrate and a green residue
C) a yellow filtrate and a brown residue
D) a green filtrate and a brown residue
- When calomel reacts with NH_4OH , we get?
A) HgNH_2Cl B) $\text{NH}_2\text{-Hg-Hg-Cl}$ C) Hg_2O D) HgO
- When phosphine is passed through aqueous solution of copper sulphate, the product produced is
A) $\text{Cu}(\text{OH})_2$ B) Cu_3P_2 C) $[\text{Cu}(\text{PH}_3)_4]^{2+}$ D) $[\text{Cu}(\text{PH}_3)_2]^{2+}$
- When an acidified solution of ferrous ammonium sulphate is treated with potassium permanganate solution, the ion which is oxidised is
A) MnO_4^- B) NH_4^+ C) Fe^{++} D) SO_4^{2-}
- Addition of $\text{K}_4\text{Fe}(\text{CN})_6$ solution to FeCl_3 solution gives
A) ferro ferricyanide B) ferri ferricyanide
C) ferric ferrocyanide D) none of these
- Anhydrous iron (III) chloride can be prepared by
A) dissolving $\text{Fe}(\text{OH})_3$ in conc. HCl
B) dissolving $\text{Fe}(\text{OH})_3$ in dil. HCl in presence of chlorine
C) passing chlorine over heated iron
D) passing dry HCl over heated iron
- When ammonia is added to a cupric salt solution, the deep blue colour so observed is due to the formation of
A) $[\text{Cu}(\text{OH})_4]^{2-}$ B) $[\text{Cu}(\text{NH}_3)_4]^{2+}$ C) $[\text{Cu}(\text{H}_2\text{O})_2(\text{NH}_3)_2]^{2+}$ D) $[\text{Cu}(\text{H}_2\text{O})_4]^{2+}$
- CrO_4^{2-} (yellow) changes to $\text{Cr}_2\text{O}_7^{2-}$ (orange) in $\text{pH} = x$ and vice-versa in $\text{pH} = y$. Hence, x and y are:
A) 6,8 B) 6,5 C) 8,6 D) 7,7
- CrO_3 dissolves in aqueous NaOH to give:
A) $\text{Cr}_2\text{O}_7^{2-}$ B) CrO_4^{2-} C) $\text{Cr}(\text{OH})_3$ D) $\text{Cr}(\text{OH})_2$
- Boiling CuCl_2 with Cu in conc. HCl gives:
A) CuCl B) CuCl_2 C) $\text{H}[\text{CuCl}_2]$ D) Cu_2Cl
- ZnO shows yellow colour on heating due to:
A) d - d transition B) C - T spectra
C) Higher polarisation caused by Zn^{2+} ion D) F - centres

13. The product of I^- with MnO_4^- in alkaline medium is:
 A) I_2 B) IO^- C) IO_3^- D) IO_4^-
14. Chrome yellow is chemically known as:
 A) lead chromate B) lead sulphate C) lead iodide D) basic lead acetate
15. On heating KMnO_4 , one among the following is not formed:
 A) K_2MnO_4 B) O_2 C) MnO_2 D) MnO
16. Standard reduction electrode potential of $\text{Zn}^{2+} / \text{Zn}$ is -0.76V . This means
 A) ZnO is reduced to Zn by H_2
 B) Zn liberates H_2 with concentrated acids
 C) Zn is generally the anode in an electrochemical cell
 D) Zn is generally the cathode in an electrochemical cell
17. Which of the following diagrams shows correct change in oxidation number of Mn?
- (A)

```

      graph TD
        MnO2 -- Increase --> MnO4_2[MnO4^2-]
        MnO2 -- decrease --> MnNH3[Mn(NH3)6]SO4
        MnO4_2 -- Increase --> MnO4_1[MnO4^-]
        MnO4_2 -- decrease --> MnNH3
      
```

(B)

```

      graph TD
        MnO2 -- Increase --> MnO4_2[MnO4^2-]
        MnO2 -- Increase --> MnNH3[Mn(NH3)6]SO4
        MnO4_2 -- decrease --> MnO4_1[MnO4^-]
        MnO4_2 -- decrease --> MnNH3
      
```

(C)

```

      graph TD
        MnO2 -- Increase --> MnO4_2[MnO4^2-]
        MnO2 -- Increase --> MnNH3[Mn(NH3)6]SO4
        MnO4_2 -- decrease --> MnO4_1[MnO4^-]
        MnO4_2 -- Increase --> MnNH3
      
```

(D)

```

      graph TD
        MnO2 -- Increase --> MnO4_2[MnO4^2-]
        MnO2 -- decrease --> MnNH3[Mn(NH3)6]SO4
        MnO4_2 -- decrease --> MnO4_1[MnO4^-]
        MnO4_2 -- Increase --> MnNH3
      
```
18. Potassium dichromate in alkaline solution, with 30% H_2O_2 produces :
 A) K_3CrO_8 B) $\text{CrO}(\text{O}_2)_2$ C) CrO_3 D) K_2CrO_4
19. Preparation of looking mirrors involves the use of:
 A) red lead
 B) ammonical silver nitrate
 C) ammonical AgNO_3 + red lead
 D) ammonical AgNO_3 + red lead + HCHO
20. A white powder soluble in NH_4OH but insoluble in water is
 A) BaSO_4 B) CuSO_4 C) PbSO_4 D) AgCl
21. Sodium thiosulphate is used to remove the unexposed AgBr from photographic films by forming a complex. In this complex of silver, the coordination number of silver is:
 A) 2 B) 3 C) 4 D) 6
22. One of the products formed due to the reaction between KMnO_4 and HCl is
 A) Red liquid, CrO_2Cl_2 B) black powder, MnO_2
 C) Greenish yellow gas, Cl_2 D) Colourless liquid, HClO_4
23. The correct statement about iron includes
 (I) The oxidation state of iron is +1 in $[\text{Fe}(\text{H}_2\text{O})_5 \text{NO}] \text{SO}_4$
 (II) That the iron shows +2 oxidation state with six electrons in the 3d orbital
 (III) The common oxidation state of iron is +3 with five unpaired electrons in the 3d

orbital

A) I, II, III

B) I, II

C) II, III

D) I only

24. The pair of the compounds in which both the metals are in the highest possible oxidation state is

A) $[\text{Fe}(\text{CN})_6]^{3-}$, $[\text{Co}(\text{CN})_6]^{3-}$

B) CrO_2Cl_2 , MnO_4^-

C) TiO_2 , MnO_2

D) $[\text{Co}(\text{CN})_6]^{3-}$, MnO_2

MULTIPLE ANSWER TYPE

25. Which of the following statements are correct?

A) An acidified solution of $\text{K}_2\text{Cr}_2\text{O}_7$ liberates iodine from iodides

B) In acidic solution dichromate ions are converted to chromate ions

C) Ammonium dichromate on heating undergoes exothermic decomposition to give Cr_2O_3

D) Potassium dichromate is used as a titrant for Fe^{2+} ions

26. When CO_2 is passed into aqueous:

A) Na_2CrO_4 solution, its yellow colour changes to orange

B) K_2MnO_4 solution, its disproportionates to KMnO_4 and MnO_2

C) $\text{Na}_2\text{Cr}_2\text{O}_7$ solution, its orange colour changes to green

D) KMnO_4 solution, its pink colour changes to green

27. Pyrolusite is MnO_2 used to prepare KMnO_4 . Steps are, $\text{MnO}_2 \xrightarrow{\text{I}} \text{MnO}_4^{2-} \xrightarrow{\text{II}} \text{MnO}_4^-$. Steps I and II are respectively:

A) fuse with KOH /air and electrolytic oxidation

B) fuse with KOH/KNO_3 and electrolytic oxidation

C) fuse with conc. HNO_3 /air and electrolytic oxidation

D) dissolve in H_2O and oxidation

28. Cuprous chloride can be prepared:

A) by passing SO_2 through the solution containing CuSO_4 and NaCl

B) by heating excess of copper with conc. HCl in presence of a little KClO_3

C) by boiling copper sulphate solution with excess of copper turnings in presence of hydrochloric acid

D) by dissolving cupric oxide or copper carbonate in conc. HCl

29. Potassium manganate (K_2MnO_4) is formed when:

A) cuprous oxide reacts with alkaline KMnO_4 solution

B) manganese dioxide is fused with K_2CO_3 in presence of KNO_3

C) formaldehyde reacts with potassium permanganate in presence of concentrated potassium hydroxide solution

D) potassium permanganate is heated with potassium hydroxide

30. Which of the following reaction(s) is/are incorrect for silver nitrate?

A) $6\text{AgNO}_3 + 3\text{I}_2 (\text{excess}) + 3\text{H}_2\text{O} \longrightarrow \text{AgIO}_3 + 5\text{AgI} + 6\text{HNO}_3$

B) $\text{AgNO}_3 (\text{excess}) + 2\text{KCN} \longrightarrow \text{K}[\text{Ag}(\text{CN})_2] + \text{KNO}_3$

C) $2\text{AgNO}_3 + 4\text{Na}_2\text{S}_2\text{O}_3 (\text{excess}) \longrightarrow 2\text{Na}_3[\text{Ag}(\text{S}_2\text{O}_3)_2] + 2\text{NaNO}_3$

D) $\text{PH}_3 + 6\text{AgNO}_3 + 3\text{H}_2\text{O} \longrightarrow 6\text{Ag} + 6\text{HNO}_3 + \text{H}_3\text{PO}_3$

31. Potassium manganate (K_2MnO_4) is formed when :

A) Potassium permanganate reacts with formaldehyde in acidic medium.

B) Manganese dioxide is fused with potassium hydroxide in air.

C) Manganese hydroxide reacts with K_2O_2 in alkaline medium.

D) Potassium permanganate is heated at 750K.

32. The hydrated cupric chloride is strongly heated. Which of the following statement(s) is/are correct for this?

- A) It is reduced to Cu_2Cl_2 .
 B) Cupric oxide is formed along with Cu_2Cl_2 .
 C) Cl_2 is liberated only but it is not liberated along with HCl .
 D) Cl_2 and HCl both are liberated.
33. Which of the following statements regarding copper salts are true?
 A) Copper (I) disproportionates to copper and copper (II) ion in aqueous solution
 B) Copper (I) can be stabilized by the formation of insoluble complex compounds such as CuCl_2^- and $\text{Cu}(\text{CN})_2^-$
 C) Copper (II) oxide is red powder.
 D) The water of crystallization of copper sulphate is five.
34. In which of the following cases, the complex ion formed will migrate toward the anode under the electric field.
 A) Excess of aqueous KCN added to CdSO_4
 B) Aqueous solution of CuSO_4 containing excess of NH_3
 C) AgBr added to hypo solution in excess
 D) A solution of AgNO_3 in aqueous KCN .
35. Which of the following statements are correct?
 A) Interstitial compounds have high melting points, higher than those of pure metals.
 B) Permanganate titrations in presence of hydrochloric acid are unsatisfactory.
 C) KMnO_4 does not act as an oxidising agent in strong alkaline medium.
 D) KMnO_4 on heating in a current of H_2 gives MnO .
36. Which of the following statements are correct?
 A) Acidic dichromate solutions on treatment with H_2O_2 gives deep blue $\text{CrO}(\text{O}_2)_2$ in presence of ether layer
 B) A deep red liquid, CrO_2Cl_2 is formed by the reaction of chromium(III) oxide with HCl in presence of conc. H_2SO_4 .
 C) $(\text{NH}_4)_2\text{Cr}_2\text{O}_7$ on heating yields green chromium(III) oxide and nitrogen gas.
 D) $\text{K}_2\text{Cr}_2\text{O}_7$ on heating with charcoal produces K_2CO_3 .
37. Select correct statements
 A) Ni(II) compounds tends to be thermodynamically more stable than Pt(II)
 B) Pt(IV) compounds are relatively more stable than Ni(IV) compounds
 C) K_2PtCl_6 exists
 D) K_2NiCl_6 exists

PASSAGE – 1

Iron forms iron halide salts by reacting the metal directly with halogen. FeI_3 does not exist. FeF_3 is white solid in spite of five unpaired electron with d^5 configuration. FeCl_3 is soluble in water and is used as a mordant in dyeing industry.

38. FeI_3 does not exist because
 A) of large size
 B) Fe^{3+} oxidizes I^- to I_2
 C) of low lattice energy
 D) Iodine is not highly electronegative enough to oxidize Fe to Fe^{3+}
39. Anhydrous FeCl_3 can be prepared by reaction of
 A) Fe with dry chlorine
 B) Fe with dil. HCl in the presence of O_2
 C) $\text{Fe}(\text{OH})_3$ with conc. HCl
 D) Fe_2O_3 with conc. HCl

MATRIX MATCHING TYPE

40. Match the reactions in Column I with nature of the reactions/type of the products in Column II.

Column - I	Column - II
A) $\text{FeSO}_4 \xrightarrow{\Delta}$	P One of the products is coloured due to charge transfer
B) $\text{Mn}^{2+} + \text{S}_2\text{O}_8^{2-} + \text{H}_2\text{O} \longrightarrow$	Q One of the products is in + VI oxidation state
C) $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 (\text{conc.}) \longrightarrow$	R Redox reaction (saturated solution)
D) $\text{N}_2\text{H}_4 + \text{CuSO}_4 \longrightarrow$	S One of the products is acidic oxide

41. Match the reactions in Column I with nature of the reactions/type of the products in Column II.

Column - I	Column - II
A) $\text{AgNO}_3 (\text{aq}) + \text{I}_2 (\text{excess}) + \text{H}_2\text{O} \longrightarrow$	P Disproportionation
B) $\text{K}_2\text{MnO}_4 (\text{aq}) + \text{CO}_2 (\text{g}) \longrightarrow$	Q Comproportionation
C) $\text{Na}_2\text{Cr}_2\text{O}_7 + \text{C} \xrightarrow{\Delta}$	R Redox
D) $\text{CuCl}_2 (\text{aq}) + \text{Cu} (\text{s}) \longrightarrow$	S One of the products is insoluble in water

42. Match the reactions in Column I with the types of products/the use of products in Column II.

Column - I	Column - II
A) Quinol + AgBr(s) $\longrightarrow$	P Turnbull's blue pigment
B) $\text{BaCl}_2 + \text{K}_2\text{Cr}_2\text{O}_7 + \text{H}_2\text{SO}_4 \longrightarrow$	Q Schwitzer's reagent
C) $\text{FeSO}_4 + \text{K}_3[\text{Fe}(\text{CN})_6] \longrightarrow$	R Rinmann's green pigment
D) $\text{Cu}(\text{OH})_2 + \text{NH}_4\text{OH} + (\text{NH}_4)_2\text{SO}_4 \longrightarrow$	S Chromyl chloride test
E) $\text{ZnO} + \text{Co}(\text{NO}_3)_2 \xrightarrow{\Delta}$	T Photography

KEY KEY

- | | | | | | | |
|----------------------------------|---------|---------|---------------------------------|----------------------------------|---------|---------|
| 1. B | 2. C | 3. A | 4. B | 5. C | 6. C | 7. C |
| 8. B | 9. A | 10. B | 11. C | 12. D | 13. C | 14. A |
| 15. D | 16. C | 17. A | 18. A | 19. D | 20. D | 21. C |
| 22. C | 23. A | 24. B | 25. ACD | 26. AB | 27. AB | 28. ABC |
| 29. ABCD | 30. AB | 31. BCD | 32. ABD | 33. ABD | 34. ACD | 35. ABD |
| 36. ACD | 37. ABC | 38. B | 39. A | 40. A-QRS ; B-PQR ; C-PQS ; D-QR | | |
| 41. A-PRS ; B-PRS ; C-RS ; D-QRS | | | 42. A-T ; B-S ; C-P ; D-Q ; E-R | | | |